SYNTHETIC GRAPH SOURCE {COUNT} $&

Before you buy or start

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Research climate group: Growing area not resolved. This grouping is not a validated recommendation for this farm.

What you reported about this farm

Water through the dry season

Not recorded. Complete the site survey and confirm water before choosing an intensive plan.

Frost on this farm

Local frost observations are not recorded. Satellite monthly averages cannot establish first and last frost dates.

Sun and drainage

Sun: not recorded.

Drainage: not recorded. Soil-test results and crop-specific needs still need checking.

Reporting-year records - not a forecast

Past reporting-year totals and months describe what was reported. They do not promise food in the coming plan.

Original eggs {year} $&

Farmer-reported annual total for 2025: 240 eggs.

Household use: 0 eggs. Sold: not recorded.

Reported harvest months: Jun, Jul. These are survey observations, not a new production forecast.

Check the recorded unit and totals before using this row.

Fish

Farmer-reported annual total for 2025: not recorded.

Household use: not recorded. Sold: not recorded.

Reported harvest months: not recorded. These are survey observations, not a new production forecast.

Check the recorded unit and totals before using this row.

Other vegetables

Farmer-reported annual total for 2025: 12 kg.

Household use: 7 kg. Sold: 3 kg.

Reported harvest months: Jan, Dec. These are survey observations, not a new production forecast.

Check the recorded unit and totals before using this row.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care

Bees: Honeybee (managed hives)

1 existing; 2 proposed structure(s). Selected enterprise: honey. No animal head count or output is assumed.

Buying advice: match breed/stock to the product, local feed, climate and care capacity; confirm with a local livestock or beekeeping adviser.

Harvest depends on the colony, local flowering and rainfall. Inspect ripeness and leave food for the colony; there is no fixed national honey season.

Time to first product: not sourced for this enterprise.

Production reference: 25 kg honey/hive/year. From a 2008 national postal survey of 112 SABIO-member beekeepers (22.4% response rate) managing an estimated 19,520 hives at 2,094 apiary sites; figure is self-reported and un-weighted. Table 7 (same doc, p.13-14) shows wide variance by region/operation-focus/size for the same year: e.g. 'All' region, honey-focused, all-size operations = 25.1-26.9 kg/hive/yr across 3 survey sites; Western Cape honey-focused = 23.6-27.9 kg/hive/yr; Western Cape pollination-focused, small operations = only 7.0-7.5 kg/hive/yr. The same document also quotes an earlier, more rigorously validated study: 'Their average reported yield is very low, at just 13.4 kilograms per hive' for Western Cape hives (Allsopp & Cherry 2004, cited p.2 of this working paper), and historical national bee-census data of 8.3 kg/hive/annum (1911-1988 average) rising to 24 kg/hive in the 1974/5 census (same doc, p.1). No single number can be called definitive; 25 kg/hive/yr is used as the document's own headline national estimate.

Check the detailed enterprise references for feed and housing. Match the ration and space to the animal stage and housing system; provide dependable clean water.

Keep hives away from playgrounds, provide fresh water and use a dry site. Ask a registered beekeeper to check safe local placement before adding a hive.

Inspect every colony regularly for signs of notifiable bee disease and Cape-bee (Capensis) infection.

Site and secure hives against theft and vandalism, which is one of the biggest challenges facing South African beekeepers.

Register with the Department before beekeeping. Registration lasts 24 months from registration or renewal and must be renewed when it expires.

A beekeeper who fails to renew registration at expiry will be deregistered and face penalties under the Act.

Beekeeping activities and hired beekeeping services require a valid registration certificate. Check the certificate before using a beekeeper.

Every beehive must be clearly and legibly marked with the beekeeper's registration number.

Keep written records of colonies, apiary locations, diseases, disease-control measures and colony losses with their reasons.

Do not move honeybees across the Cape bee line, or keep Cape bees north of it, without written approval.

Report any suspected notifiable disease (American foulbrood is listed); officials can order an infected colony destroyed.

Source: FAO: Beekeeping in Africa, Site selection (general African apiary guidance)

Source: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, p. 6

Source: Gums & Bees: A roadmap for landowners in South Africa (South African National Biodiversity Institute, SANBI), p. 21

Source: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, control measures 2(1) and 2(4), p. 4

Source: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, control measure 2(5), p. 4

Source: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, control measures 2(8) and 2(9), p. 5

Source: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, p. 5

Source: Survey of Beekeeping in South Africa, CSSR Working Paper No. 221 (Conradie & Nortje, 2008, UCT Centre for Social Science Research), p. 14

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Bees: Honeybee (managed hives) (continued)

Source: FAO, Beekeeping in Africa, honey harvesting

Kraal: Nguni / crossbred beef cattle (communal and extensive veld)

1 existing; 2 proposed structure(s). Selected enterprise: meat. No animal head count or output is assumed.

Buying advice: match breed/stock to the product, local feed, climate and care capacity; confirm with a local livestock or beekeeping adviser.

Time to first product: not sourced for this enterprise.

Production reference: 114.5-121.2 kg weaning weight/calf. Table 5 gives weaning weight (WW) across four agro-ecological zones for the same nine Nguni communal herds: arid 114.51 kg (lowest), semi-arid 119.98 kg, dry sub-humid 117.91 kg, humid 121.21 kg (highest). The range recorded here spans this one study's own arid-to-humid zone means; it is not merged with any other paper's weaning-weight figures (other SA studies report different ranges for different systems - e.g. research-station or stud herds - and were deliberately excluded to avoid mixing systems). 826 Nguni calves from nine herds that the paper itself says were 'kept under extensive grazing system' in Limpopo Province - a genuine communal/extensive veld system matching this enterprise.

Productive-life reference: 6.4 years. 2354 days / 365.25 = 6.44 years. 'Failure time' is the length of productive herd life for the 580 uncensored (fully-observed) records in the combined dataset; the dissertation itself defines productive herd life as 'the time interval between first calving date to culling date'. Combined data from three extensively-grazed Nguni stud herds, animals born 1968-2015.

Check the detailed enterprise references for feed and housing. Match the ration and space to the animal stage and housing system; provide dependable clean water.

Cattle pens must not be overcrowded: there must be enough floor space for every animal to lie down at the same time.

During transport, cattle must have at least 1.4 m² of floor space each so loading is neither too tight nor too loose.

Cattle may not be transported for more than 18 hours without being off-loaded and given feed, water and shelter.

It is an offence to starve, under-feed, or deny water or food to cattle.

Foot-and-mouth disease (FMD) is a controlled animal disease under the Animal Diseases Act: state Veterinary Services enforce isolation and movement restrictions.

Every cattle owner must register an official identification mark and brand or mark their cattle with it, under the Animal Identification Act.

Any suspected or confirmed brucellosis-related abortion in cattle must be reported to the State Veterinary office, as part of the national bovine brucellosis control scheme under the Animal Diseases Act.

No animal may be slaughtered anywhere but a registered abattoir, with the exception below.

Slaughter for the farmer's own household consumption, or for cultural or religious purposes, is exempt from the abattoir rule.

Source: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.1.2, p. 4

Source: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.4.6.1, p. 7

Source: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.7.1, p. 9

Source: Animals Protection Act 71 of 1962, s.2(1)(c), p. 1

Source: SAnews: KZN Foot and Mouth Disease boundaries extended (Dept of Agriculture statement)

Source: Animal Identification Act 6 of 2002, s.7(1), p. 4

Source: Marumo, Hlokwe & Kayoka-Kabongo, Seroprevalence of brucellosis in communal and smallholder cattle farming in North West Province, South Africa, Onderstepoort Journal of Veterinary Research (PMC10784229)

Source: Meat Safety Act 40 of 2000, s.7(1)(a), p. 6

Source: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Kraal: Nguni / crossbred beef cattle (communal and extensive veld) (continued)

Source: Mpofu, Ginindza, Siwendu, Nephawe & Mtileni (2016). Effect of agro-ecological zone, season of birth and sex on pre-weaning performance of Nguni calves in Limpopo Province, South Africa. Tropical Animal Health and Production 49:187-194 (PMC5203849), Table 5

Source: Ngayo, M. (2017). Factors Affecting Productive Life and Fertility in Nguni Cows (MSc dissertation, University of the Free State), Table 5.1, p. 58

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Kraal: Small-scale dairy cow (Jersey, Holstein or crossbred)

1 existing; 2 proposed structure(s). Selected enterprise: milk. No animal head count or output is assumed.

Buying advice: match breed/stock to the product, local feed, climate and care capacity; confirm with a local livestock or beekeeping adviser.

Plan around actual calving/kidding and lactation. Dry animals do not supply milk.

First product reference: 121.8-130.5 weeks. Table 1, AFC column, rows in the table's order Holstein-Friesland Reg./Grade, Jersey Reg./Grade, Ayrshire Reg./Grade: 28, 30, 25, 28, 29, 32 months. The Grade rows used here are Holstein-Friesland 30 and Jersey 28 (Grade Ayrshire 32 is not used: the enterprise is Jersey or Holstein type). 28 x 4.35 = 121.8 weeks; 30 x 4.35 = 130.5 weeks. Milk begins at first calving, so age at first calving is the time to first product.

Production reference: 3840-5422 kg milk/cow/lactation. Table 1 ('Comparison of number of cows, production and fertility in the main dairy breeds in South Africa'), Milk column, Grade rows: Jersey 3840 kg and Holstein-Friesland 5422 kg per lactation (Registered rows: 4468 and 6426 kg). The document says these are National Milk Recording Scheme herds, which 'have been shown to be better managed than those outside the scheme, with their performance per lactation being 50 % greater than that of the national herd', so a smallholder herd should expect less.

Productive-life reference: 1.9-2 years. Holstein 739.33 and Jersey 696.81 days of productive life (first calving to culling) in South African milk-recorded herds, 2000-2019: 696.81/365.25 = 1.91 and 739.33/365.25 = 2.02 years.

Age/system reference: In hotter, humid KZN, KZN DARD describes Nguni × Jersey dairy trials at Makhathini to combine milk potential with local adaptation. Hand milking with the calf present highlights competition between household milk and calf growth. Discuss adapted stock and feeding locally; commercial milk-recording totals do not forecast this system. Record calving, lactation stage and milk available after calf needs.

Check the detailed enterprise references for feed and housing. Match the ration and space to the animal stage and housing system; provide dependable clean water.

Cattle pens must not be overcrowded: there must be enough floor space for every animal to lie down at the same time.

Cows must not be loaded for transport if their udders are distended with milk.

Cattle may not be transported for more than 18 hours without being off-loaded and given feed, water and shelter.

It is an offence to starve, under-feed, or deny water or food to a dairy cow.

Cows on pasture or in open camps need shade: shade structures should give at least 4 to 5 m² of shade per cow.

Foot-and-mouth disease (FMD) is a controlled animal disease under the Animal Diseases Act: state Veterinary Services enforce isolation and movement restrictions.

Every cattle owner must register an official identification mark and brand or mark their cattle with it, under the Animal Identification Act.

Dairy farmers are legally required to test their herds for bovine TB, mainly because of export requirements, so most dairy cattle in South Africa are tested.

Any suspected or confirmed brucellosis-related abortion in cattle must be reported to the State Veterinary office, under the national bovine brucellosis control scheme.

A cull cow may be slaughtered only at an abattoir, except for the farmer's own consumption or for cultural or religious purposes.

Source: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.1.2, p. 4

Source: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.6.13, p. 9

Source: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.7.1, p. 9

Source: Animals Protection Act 71 of 1962, s.2(1)(c), p. 1

Source: Muller, C.J.C., The Dairy Farming Handbook (2017), Western Cape Department of Agriculture (Elsenburg), p. 129

Source: SAnews: KZN Foot and Mouth Disease boundaries extended (Dept of Agriculture statement)

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Kraal: Small-scale dairy cow (Jersey, Holstein or crossbred) (continued)

Source: Animal Identification Act 6 of 2002, s.7(1), p. 4

Source: Bovine tuberculosis / dairy herd testing study, PMC10369683

Source: Marumo, Hlokwe & Kayoka-Kabongo, Seroprevalence of brucellosis in communal and smallholder cattle farming in North West Province, South Africa, Onderstepoort Journal of Veterinary Research (PMC10784229)

Source: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Source: KZN DARD: Subsistence Dairying in KZN, Makhathini Research Station trials

Source: Muller, C.J.C., The Dairy Farming Handbook (2017), Western Cape Department of Agriculture (Elsenburg), p. 10

Source: Muller, C.J.C., The Dairy Farming Handbook (2017), Western Cape Department of Agriculture (Elsenburg), Table 21.1, p. 122

Source: Gertenbach, W.D., Dairying in KwaZulu-Natal: Breeds of Dairy Cattle, Cedara Agricultural Development Institute, KZN Department of Agriculture and Rural Development, Table 1, p. 1

Source: Cow Longevity and Reasons and Risk Factors for Culling in South African Holstein and Jersey Dairy Herds, Animals 2025 (PMC12560872)

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Chickens: Broilers (commercial)

1 existing; 2 proposed structure(s). Selected enterprise: meat. No animal head count or output is assumed.

Buying advice: choose a meat strain and supplier batch plan; confirm brooding, feed, water and slaughter arrangements.

Plan balanced feed, clean water, shade, ventilation and predator protection. Egg supply depends on laying birds, their age, health and care; a coop alone supplies no eggs.

First product reference: 4.6-4.7 weeks. 32 days / 7 = 4.57 weeks; 33 days / 7 = 4.71 weeks. This enterprise has a single grow-out cycle ending in slaughter, so 'time to first product' = age at slaughter. South Africa, commercial industry average.

Production reference: 1.8-1.9 kg/bird at slaughter (32-33 days). National commercial broiler industry average, South Africa, 2022. A separate breed-trial table (Table 4b, same document, p.39) gives named-strain 'Cumulative FI (kg)' and 'Body weight (kg)' at exactly 35 days: Arbor Acres Plus 2.29kg, Cobb 500 2.52kg, Indian River 2.30kg, Ross 308 2.30kg (cumulative feed intake 3.21-3.64kg) - a slightly older age (35 vs 32-33 days) and higher weight than the industry-wide average quoted above; not merged into the primary figure since it is a different (breed-standard trial) dataset, but used below for the feed-intake figure.

Age/system reference: Newer South African commercial reference: SAPA 2024 reports average slaughter at 30-31 days with bird weight 1.8-1.85 kg. The primary range above remains the earlier 2022 reference at 32-33 days. These are commercial bird weights, not dressed meat kg or a village flock target. Record hatch or purchase age, feeding system, actual harvest date and weighed product basis.

Check the detailed enterprise references for feed and housing. Match the ration and space to the animal stage and housing system; provide dependable clean water.

Floor-housed broilers: no more than 40 kg of bird per square metre.

Newly hatched chicks must be given feed within 24 hours of hatching.

Newly hatched chicks must be given access to water within 24 hours of hatching, sooner in hot weather.

Fresh water and the right feed always available, so birds are free from hunger and thirst.

Broiler houses need a minimum ventilation/airflow rate of 5 cubic metres per hour for every kilogram of bird mass in the house.

In free-range broiler production, indoor stocking must not exceed 15 birds per square metre of floor space.

An owner or manager whose animals are, or may be, infected with a controlled animal disease must report it to the state veterinary director immediately.

Newcastle disease and notifiable avian influenza are on the state list of controlled and notifiable animal diseases.

No animal may be slaughtered anywhere but an abattoir, with the exceptions below.

Slaughter for your own household, or for cultural or religious purposes, is exempt from the abattoir rule.

Source: SAPA Code of Practice for Broiler Production (May 2012), 2.2 Housing of broilers in Floor Systems, p. 6

Source: SAPA Code of Practice for Broiler Production (May 2012), p. 11

Source: SAPA Code of Practice for Broiler Production (May 2012), p. 12

Source: SAPA Code of Practice for Broiler Production (May 2012), p. 7

Source: SAPA Code of Practice for Broiler Production (May 2012), p. 8

Source: SAPA Code of Practice for Broiler Production (May 2012), 2.4 Free Range Broiler Production, p. 8

Source: Animal Diseases Act 35 of 1984, s.11(1)(b)(ii), p. 9

Source: Department of Agriculture, Forestry and Fisheries, Controlled and Notifiable Animal Diseases (factsheet), p. 1

Source: Meat Safety Act 40 of 2000, s.7(1)(a), p. 6

Source: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Source: SAPA Industry Profile 2024, section 7.9 Performance efficiency, p. 82

Source: SAPA Industry Profile 2022, section 6.9 Performance efficiency, p. 79

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Chickens: Broilers (commercial) (continued)

Source: ARC Climate-Smart Agriculture, module 11, printed pp. 468-480

Chickens: Village chickens (indigenous, free-range)

1 existing; 2 proposed structure(s). Selected enterprise: eggs. No animal head count or output is assumed.

Buying advice: Potchefstroom Koekoek is a dual-purpose option; Venda, Ovambo and Naked Neck are other ARC-conserved options. No province-specific winning breed is claimed.

Plan balanced feed, clean water, shade, ventilation and predator protection. Egg supply depends on laying birds, their age, health and care; a coop alone supplies no eggs.

Collect eggs frequently, keep nest litter clean and dry, and handle/store eggs carefully for household use or sale.

First product reference: 15.7-27.8 weeks. Free-range village chickens, KwaZulu-Natal; dry-environment birds started laying earlier (15.74 weeks) than wet-environment birds (27.78 weeks), difference flagged significant ('*') in the source table. Wide range reflects genuine dry/wet-season environmental variation reported by the source, not measurement uncertainty.

Production reference: 40.4-49.9 eggs/hen/year. Free-range/scavenging village chickens, KwaZulu-Natal: dry (40.43) vs wet (49.91) environment, difference not significant ('NS'). A managed-system ARC Irene trial of named indigenous breeds (Grobbelaar, Sutherland & Molalakgotla 2010, Animal Genetic Resources 46: 25-32) reports far higher counts, now separately verified and retained in productionReferences; managed conditions cannot replace this village range.

Age/system reference: The ARC Irene trial compared Potchefstroom Koekoek, Venda, Ovambo and Naked Neck under commercial feed, water and controlled lighting. Mean eggs/hen in a 52-week cycle were 195.9, 153.7, 125.5 and 138.9 respectively; first-egg ages were 138.5, 139.0, 134.5 and 129.0 days from hatch, with no significant first-egg age difference among breeds. This is managed breed context, not the yield of scavenging hens in KZN. Match feeding and housing before using a benchmark.

Check the detailed enterprise references for feed and housing. Match the ration and space to the animal stage and housing system; provide dependable clean water.

In extensive village systems the chickens have to find their own water and food.

Scavenging village chickens grow more slowly than commercially-farmed birds and are more vulnerable to disease, parasites and predators.

In the Limpopo and KwaZulu-Natal survey, 94.9% of farmers housed their chickens at night.

What free-ranging hens forage met their needs for maintenance and growth but not for laying: protein in every season, calcium and phosphorus in winter.

An owner or manager whose animals are, or may be, infected with a controlled animal disease must report it to the state veterinary director immediately.

Newcastle disease and notifiable avian influenza are on the state list of controlled and notifiable animal diseases.

No animal may be slaughtered anywhere but an abattoir, with the exceptions below.

Slaughter for your own household, or for cultural or religious purposes, is exempt from the abattoir rule.

Source: ARC, Climate-Smart Agriculture Training Manual, Module 11: Poultry Production, p. 471

Source: ARC, Climate-Smart Agriculture Training Manual, Module 11: Poultry Production, p. 468

Source: Malatji, Tsotetsi, van Marle-Koster & Muchadeyi (2016), 'A description of village chicken production systems and prevalence of gastrointestinal parasites: Case studies in Limpopo and KwaZulu-Natal provinces of South Africa', Onderstepoort Journal of Veterinary Research 83(1)

Source: Raphulu, Jansen van Rensburg & van Ryssen (2015), 'Nutrient content of crop contents of free ranging indigenous chickens raised under village conditions', South African Journal of Animal Science 45(2), p. 1

Source: Animal Diseases Act 35 of 1984, s.11(1)(b)(ii), p. 9

Source: Department of Agriculture, Forestry and Fisheries, Controlled and Notifiable Animal Diseases (factsheet), p. 1

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Chickens: Village chickens (indigenous, free-range) (continued)

Source: Meat Safety Act 40 of 2000, s.7(1)(a), p. 6

Source: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Source: Grobbelaar, Sutherland & Molalakgotla (2010), Egg production potentials of certain indigenous chicken breeds from South Africa: methods pp27-28, first-egg p29, Table 4 p31, p. 31

Source: Tenza, Mhlongo & Chimonyo (2023), 'Village chicken production and egg quality in dry and wet, resource-limited environments in KwaZulu-Natal, South Africa', South African Journal of Animal Science 53(6), p. 5

Source: ARC Climate-Smart Agriculture, module 11, printed pp. 468-480

Source: FAO: Small-scale poultry production, egg marketing

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Chickens: Laying hens (commercial)

1 existing; 2 proposed structure(s). Selected enterprise: eggs. No animal head count or output is assumed.

Buying advice: for egg production, request vaccinated point-of-lay pullets from a layer supplier, with age, feeding and vaccination records.

Plan balanced feed, clean water, shade, ventilation and predator protection. Egg supply depends on laying birds, their age, health and care; a coop alone supplies no eggs.

Collect eggs frequently, keep nest litter clean and dry, and handle/store eggs carefully for household use or sale.

Season reference: With a constant 14-16 hour light day, commercial hens keep laying at a high rate all year. This commercial management reference is not a naturally flat household egg supply.

First product reference: 21 weeks. Bird age from hatch; confirm the strain and rearing with the supplier. Point-of-lay birds start later in this cycle.

Production reference: 407.1-428.6 eggs/hen housed, laying cycle to 90 weeks. Table 4a's cumulative 'Eggs/hen housed' for five commercial strains; the same table gives 'Age at depop. weeks 90.0' for every strain. Commercial barn/cage average, not a free-range smallholder trial. Kept as the source gives it: a per-year rate would need a point-of-lay age from another document.

Age/system reference: SAPA 2024 commercial strain standards cumulative to 90 weeks: Dekalb Amberlink 429 eggs/hen housed, Dekalb Brown 428, Hy-Line Silver Brown 435, Hy-Line Brown 434 and Lohmann Brown Lite 413. These are full-cycle objectives, not eggs per calendar year or a smallholder forecast. Purchase age, actual onset of lay, surviving laying hens, feed, water and lighting need separate records.

Check the detailed enterprise references for feed and housing. Match the ration and space to the animal stage and housing system; provide dependable clean water.

In free-range systems, hens must have access to the outdoor range for at least 8 hours a day in daylight.

Where individual nest boxes are used, at least one nest for every 8 hens.

Hens need at least 8 hours of continuous darkness in every 24-hour cycle to rest properly.

Floor housing needs litter on at least a third (33%) of the floor.

The outdoor free-range area must not be overstocked: no more than 5 birds per square metre of range.

An owner or manager whose animals are, or may be, infected with a controlled animal disease must report it to the state veterinary director immediately.

Newcastle disease and notifiable avian influenza are on the state list of controlled and notifiable animal diseases.

No animal may be slaughtered anywhere but an abattoir, with the exceptions below.

Slaughter for your own household, or for cultural or religious purposes, is exempt from the abattoir rule.

Source: SAPA Abridged Code of Practice: Commercial Layers, p. 1

Source: Animal Diseases Act 35 of 1984, s.11(1)(b)(ii), p. 9

Source: Department of Agriculture, Forestry and Fisheries, Controlled and Notifiable Animal Diseases (factsheet), p. 1

Source: Meat Safety Act 40 of 2000, s.7(1)(a), p. 6

Source: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Source: SAPA Industry Profile 2024, Table 4: commercial standards cumulative to 90 weeks, p. 39

Source: SAPA Layer Notes, Commercial Layers (training module, November 2013), p. 13

Source: SAPA Industry Profile 2022, Table 4a (Breed standards in laying hens), p. 39

Source: ARC Climate-Smart Agriculture Training Manual, Poultry production, p. 8

Source: ARC Climate-Smart Agriculture, module 11, printed pp. 468-480

Source: FAO: Small-scale poultry production, egg marketing

Source: SAPA Layer Notes, Commercial Layers (training module, November 2013)

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Ducks: Meat duck (Pekin / Muscovy, backyard)

1 existing; 2 proposed structure(s). Selected enterprise: meat. No animal head count or output is assumed.

Buying advice: match breed/stock to the product, local feed, climate and care capacity; confirm with a local livestock or beekeeping adviser.

First product reference: 7-12 weeks. P44 = Pekin (7 weeks to slaughter); MR71 = Muscovy (10 weeks for ducks/hens, 12 weeks for drakes). Treated as 'time to first product' = age to slaughter, since this is a single-cycle meat enterprise (no repeat product like eggs). Non-South-African trial by Warsaw researchers, published in SAJAS; journal location does not establish local applicability.

Production reference: 2.3-2.9 kg live weight/bird at slaughter. Non-SA (Guyana study reporting a citation to Steklenev 1990); no SA-specific absolute slaughter live-weight figure was found (see gaps). 2.93 kg = Pekin at 8-10 weeks, 2.29 kg = Muscovy at 8-10 weeks - matches this enterprise's two breeds. The same LRRD paper's own experiment (Table 2, intensive system, Guyana) measured live weight at slaughter (8 weeks) as: Pekin drakes 2.53 kg, Pekin hens 2.29 kg, Muscovy drakes 2.47 kg, Muscovy hens 1.51 kg - broadly consistent with, but not merged into, the cited range above since it is a separate dataset/table in the same source. No sourced eggs/duck/year figure was found for either breed (see gaps); the LRRD and SAJAS sources used here studied ducks reared only for meat.

Check the detailed enterprise references for feed and housing. Match the ration and space to the animal stage and housing system; provide dependable clean water.

It is an offence to deny a duck food or water, or keep it somewhere lacking adequate space, ventilation, light or shelter from heat and cold.

Watch flocks for sudden deaths, ruffled feathers, swollen wattles/combs or a sudden drop in egg production - these are signs of avian influenza and should be reported at once.

Keep ducks away from wild birds and open water/feed that attracts them, and control visitors' access to duck housing, to reduce avian-influenza risk.

Ducks are one of the domesticated animal species listed in the Meat Safety Act, so its rules on registered abattoirs and hygiene apply to them.

Ducks slaughtered to be sold must be slaughtered at a registered abattoir; slaughter for the farmer's own consumption is exempt but that meat may not be sold.

At the abattoir, ducks must be handled humanely during loading, transport, offloading, housing, immobilising and killing, under the Animals Protection Act.

Ducks are birds/poultry, so an outbreak of avian influenza in a duck flock is a controlled animal disease under the Animal Diseases Act and must be reported to the state vet immediately.

Source: Animals Protection Act 71 of 1962 (consolidated text, s.2(1)(b)-(c)), p. 2

Source: Western Cape Department of Agriculture, Avian Influenza (Bird Flu) factsheet, p. 2

Source: Meat Safety Act 40 of 2000, Schedule 1 (Animals to which this Act applies), p. 16

Source: Meat Safety Act 40 of 2000, s.7, p. 6

Source: Meat Safety Act 40 of 2000, s.9, p. 7

Source: Damaziak et al. (2014), Influence of housing system on the growth and histological structure of duck muscles, S. Afr. J. Anim. Sci. 44(2): Poland trial

Source: Livestock Research for Rural Development 18(11) 2006 - Solomon et al. (Guyana; not an SA figure), citing Steklenev 1990

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Animal products and care (continued)

Pond: Tilapia (Mozambique tilapia, small pond or tank)

1 existing; 2 proposed structure(s). Selected enterprise: fish. No animal head count or output is assumed.

Buying advice: match breed/stock to the product, local feed, climate and care capacity; confirm with a local livestock or beekeeping adviser.

First product reference: 43.5-52.2 weeks. 10 months x 4.35 = 43.5 weeks; 12 months x 4.35 = 52.2 weeks. This is Mozambique Tilapia (O. mossambicus) grow-out time to harvestable size; the source separately states Nile Tilapia (O. niloticus) reaches harvestable maturity faster, in 6-10 months (= 26.1-43.5 weeks) - not used here as it is a different species with a different legal status (see 'legal').

Production reference: 800 g harvest weight/fish. Table 8-3 gives the Mozambique Tilapia's harvestable size as an approximate single figure ('+- 800g'), not a range, so [800, 800] is used rather than inventing a spread. The same table cell wraps '10-12' across a line break in the PDF ('10-\n12'), kept verbatim. The table's Nile Tilapia row gives the same approximate weight (+-800g) reached faster, in 6-10 months - not used here because this dossier is framed around the indigenous, legally unrestricted Mozambique Tilapia per task instruction; see 'legal' for why Nile Tilapia is treated separately. No SA source was found giving a feed-conversion-based or per-hectare yield figure specific to O. mossambicus that could be converted to this per-fish unit without stitching numbers across documents, so outputUnit follows the source's own unit (grams harvest weight per fish) rather than being converted.

Age/system reference: ARC makes tilapia suitability depend on water temperature: summer grow-out can be suitable anywhere in South Africa when water exceeds 20°C; growth is poor below 18°C and winter water below 13°C can kill farmed tilapia. The Highveld reference is a summer crop; tunnels can widen the range. Measure pond water and confirm stocking species and dates. Air climate or a mapped pond does not prove harvest months, fish numbers or food kg.

Check the detailed enterprise references for feed and housing. Match the ration and space to the animal stage and housing system; provide dependable clean water.

The indigenous Mozambique Tilapia (O. mossambicus) is not a listed invasive species and may be farmed without an alien-species permit; the faster-growing Nile Tilapia (O. niloticus) is.

Nile Tilapia (Oreochromis niloticus) is a nationally listed invasive fish species: Category 1b inside National Parks, Provincial Reserves, Mountain Catchment Areas and Forestry Reserves, and Category 3 in all other areas.

Outside protected areas Nile tilapia is Category 3: it may remain in prescribed areas or provinces, but further propagation or trade is prohibited.

Nile Tilapia has been listed under the NEMBA Alien and Invasive Species Regulations as requiring a permit to produce.

Producing an alien or invasive species (such as Nile Tilapia) is prohibited in areas designated under the Protected Areas Act and the Fish Sanctuary Act; a permit is required to produce it elsewhere.

Producing tilapia in South Africa requires a water use licence under the National Water Act.

Source: IDC, Research into the Potential for the Production, Processing and Export of Tilapia for the Southern African Market (2015), Table 8-3 (Mozambique vs. Nile Tilapia), p. 69

Source: Dept. of Environmental Affairs / invasives.org.za, National Listed Invasive Species (NEMBA AIS Regulations booklet), List 7 (National List of Invasive Fresh-water Fish Species), p. 32

Source: Dept. of Environmental Affairs / invasives.org.za, National Listed Invasive Species (NEMBA AIS Regulations booklet), p. 3

Source: IDC, Research into the Potential for the Production, Processing and Export of Tilapia for the Southern African Market (2015), p. 69

Source: IDC, Research into the Potential for the Production, Processing and Export of Tilapia for the Southern African Market (2015), p. 74

Source: IDC, Research into the Potential for the Production, Processing and Export of Tilapia for the Southern African Market (2015), p. 75

Source: ARC Aquaculture Production App: General FAQ, Can I grow tilapia on the Highveld?

Crop varieties to check locally

Keep the named variety you actually choose. Research shortlists need confirmation for your local season, water and ground.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Farm choices - continued

Use your farm observations and the research shortlists. Confirm choices locally before buying.

Crop varieties to check locally (continued)

Tomatoes

Recorded variety: not chosen yet.

This site's climate has not resolved, so no local shortlist is claimed. Open the crop card to see sources and record your choice.